

AI PDF Assistant

LLM-Powered Document Analysis — Project Overview

1. Introduction

AI PDF Assistant is a document-analysis application designed to make long PDF files easier to understand. Users can upload a PDF, extract its text, generate a concise summary, and ask questions about the document using a large language model.

2. Core Workflow

The application follows a simple pipeline: PDF upload → text extraction → document processing → LLM request → generated response. The interface is designed to keep the workflow accessible while providing useful answers grounded in the uploaded document.

3. Main Features

PDF Upload: Accepts user-provided PDF documents.

Text Extraction: Converts document content into text that can be processed by the language model.

Summarization: Produces a concise overview of important content.

Question Answering: Allows users to ask context-aware questions about the document.

4. Technology Stack

The project can be implemented with Python, Streamlit, PyPDF/PyPDF2-style PDF processing libraries, and the Groq API. The LLM layer is responsible for summarization and natural-language question answering.

5. Typical Use Cases

Students can use the system to understand lecture notes and reports. Developers can analyze technical documentation. Researchers can quickly review lengthy papers. Professionals can extract important information from business documents.

6. Limitations

Generated answers depend on the quality and completeness of extracted PDF text and the model's context. Scanned PDFs may require OCR before their content can be processed reliably. Users should verify important information against the original document.

This document is provided as downloadable project/reference material.